Answer key: Lesson 9, Spot the Mistake

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. A child writes 41 – 17 = 24.

Answer: 41; Yes

24 + 17 = 41. Adding back gives the number we started with, so the answer is right.

Watch for: Writing 24 + 17 = 31 (a counting slip).; Saying No without adding back.

S2. A child writes 50 – 23 = 37.

Answer: 60; No; 27

37 + 23 = 60, not 50. So 37 is wrong. 27 + 23 = 50, so the right answer is 27.

Watch for: Saying Yes because 37 looks about right.; Writing 73 (adding 50 and 23).

S3. A child adds 36 + 28 in three steps. One step is wrong.

Answer: Step 2; 64

Step 1: 36 + 4 = 40, right. Step 2: 40 + 20 = 60, not 50. Step 3 then starts from 60: 60 + 4 = 64. The right answer is 64. Check: 64 – 28 = 36.

Watch for: Circling step 3 because it is the last step.; Writing 54 (the child's answer).

S4. Ravi starts to read at the time on the clock. He reads for 1 hour. A child says he stops at 3:00.

Answer: No; 3:30

It starts at half past 2. One hour later the short hand moves to the next number and the long hand stays on the 6. That is half past 3, or 3:30. The child forgot the half hour.

Watch for: Saying Yes because the hour changed.; Writing 3:00 or 2:30.

S5. A bag costs $7 and a cup costs $5. A child says they cost $10 altogether.

Answer: No; $12

7 and 5: 7 and 3 make 10, and 2 more make 12. They cost $12. Check: 10 take away 7 is 3, and the cup costs 5, so 10 is wrong.

Watch for: Saying Yes because 10 is close.; Taking away: 7 take away 5 is 2.

S6. Tick True or False for each sum.

Answer: True; False; True; False

25 + 30 = 55 is true. 56 – 20 = 36, not 46, so it is false. 24 + 9 = 33 is true. 80 – 5 = 75, not 85, so it is false.

Watch for: Ticking True for every sum.; Taking away the tens digit by mistake: 56 – 20 = 46.

Think harder (page 3)

T1. A child works out 74 – 29 in two steps. Step 1 takes away 30 on purpose, because 29 is close to 30. That is a good first step.

Answer: Step 2; 45

Step 1: 74 – 30 = 44. Taking away 30 is a good first step, but 29 is 1 less than 30, so we took away 1 too many. Step 2 must add 1 back: 44 + 1 = 45. The child took away another 1 instead. Check: 45 + 29 = 74.

Watch for: Circling step 1: taking away 30 is allowed, it only takes away 1 too many.; Writing 44 (stopping after step 1).

T2. Two children work out 58 + 27. One child writes 75. The other child writes 85.

Answer: 48; 58; 85

75 – 27 = 48, not 58. 85 – 27 = 58, which is where we started. So 85 is right.

Watch for: Choosing 75 because 5 + 7 is 12 and the ten is forgotten.; Adding instead of taking away when checking.

T3. Mei has 28 pencils. Mei has 6 fewer pencils than Aisha. A child writes 28 – 6 = 22 to find how many pencils Aisha has.

Answer: No; 34

Mei has fewer, so Aisha has more. Aisha has 6 more than 28: 28 + 6 = 34. Check: 34 has 6 more than 28, and 28 has 6 fewer than 34.

Watch for: Saying Yes because fewer means take away.; Writing 22.

T4. Ravi leaves home at 7:30. He walks for half an hour. A child says he gets there at 7:60. Look at the clocks.

Answer: B; 8:00

A half hour after half past 7 is 8 o'clock. There is no 7:60 on a clock: after 7:59 comes 8:00. Clock B shows 8:00.

Watch for: Choosing clock A (7:00) or C (8:30).; Writing 7:60.

Go further (page 4)

G1. A child writes 52 + 39 = 191.

Answer: about 90; No; 91

52 is near 50 and 39 is near 40, and 50 and 40 make 90. So the answer is about 90. 191 is much too big, so it cannot be right. 52 + 39 = 91.

Watch for: Guessing about 150 because both numbers have a 9 or a 5.; Saying Yes because 191 looks like 52 and 39 joined.

G2. A child adds 47 + 38 in columns and writes 75. One digit is wrong.

Answer: Tens; 85

Ones: 7 + 8 = 15. Write 5 and carry 1 ten. Tens: 4 + 3 + the 1 carried ten = 8. The child forgot the carried ten, so the tens digit should be 8. The right answer is 85.

Watch for: Circling Ones because 7 + 8 looks hard.; Writing 75 again.

G3. Write a sum with one mistake for a grown-up to find. Many answers are right.

Answer: Many answers are right: the sum has exactly one mistake and the right answer is given

For example 38 + 15 = 43: the right answer is 53, and the mistake is that the ten was forgotten. Check: add back 43 – 15 = 28, not 38.

Watch for: Writing a sum with no mistake.; Changing two things at once.

Investigation (page 5): The same mistake three times

First: 34 – 19 = 15, not 25, so it is wrong. The right answer is 44. Each wrong answer is 10 less than the right one (44, 63 and 83). The child forgets the ten made by adding the ones. Another sum the child gets wrong needs 10 or more in the ones, for example 29 + 14 = 33 (right: 43).

Look for: A child who finds that every wrong answer is exactly 10 less than the right one, or who says the child forgets the carried ten. Credit any clear reason. A new sum must make 10 or more in the ones; a sum with no carry is not a good example.

Exit ticket (page 5)

E1. A child writes 82 – 47 = 45.

Answer: 92; No

45 + 47 = 92, not 82. So 45 is wrong. The right answer is 35, because 35 + 47 = 82.

Watch for: Saying Yes because 45 looks right.; Writing 82 for part a.

E2. A child works out 28 + 15 in steps. Which step is wrong? Write its number in the bracket.

Answer: (2) 15 – 2 = 12

15 split into 2 and 13. 15 – 2 = 13, not 12. Then 30 + 13 = 43.

Watch for: Choosing (3) because it is the last step.; Choosing (1) because 28 + 2 looks odd.

E3. Lily has $20. She spends $12. A child says she has $9 left.

Answer: No; $8

Count up from 12 to 20: 8. Check: 8 and 12 make 20. The child's 9 and 12 make 21, so 9 is wrong.

Watch for: Saying Yes.; Adding 20 and 12.

Sneak-peek puzzles (not in the workbook)

1. No. 40 + 20 = 60.
2. No. 27 + 16 = 43. The child forgot the ten: 7 + 6 = 13, which is 1 ten and 3 ones. A check: 33 take away 16 is 17, not 27.